

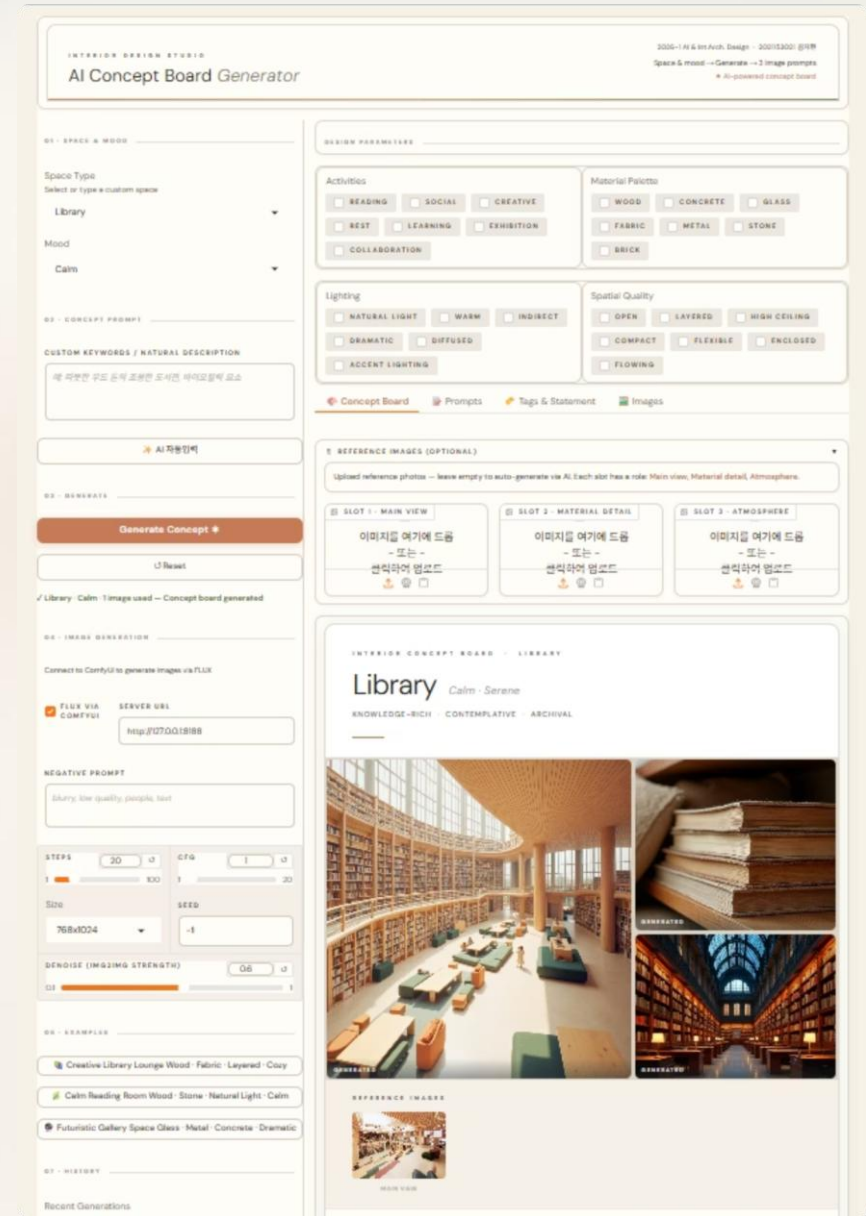
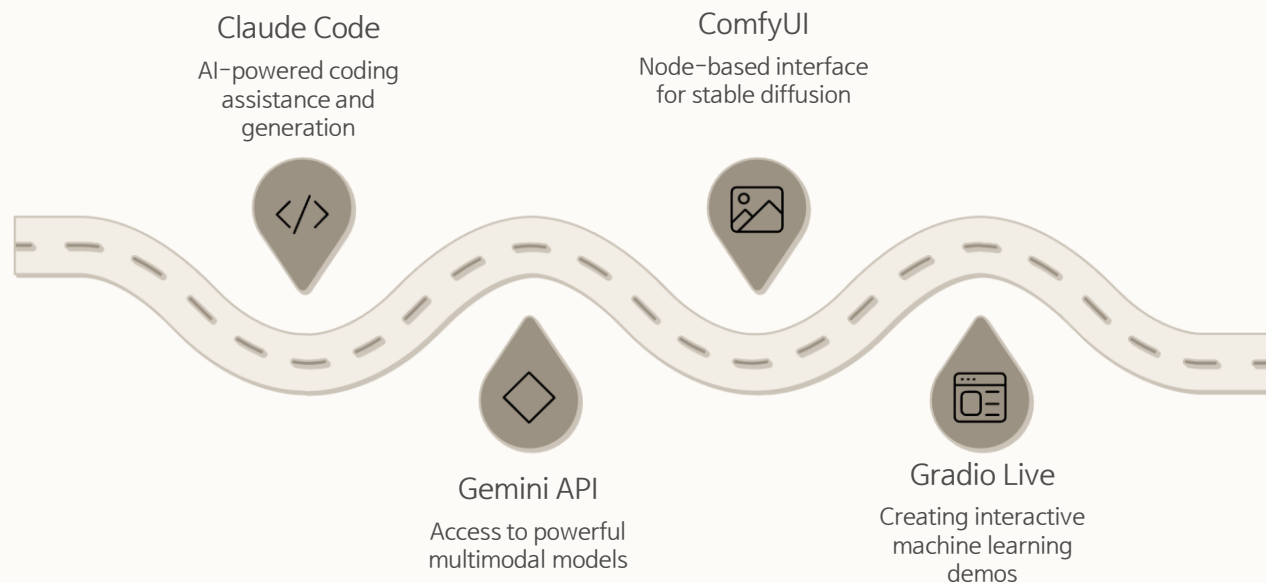
AI Interior Concept Board Generator

Developed with Claude Code | Runtime: Gemini API + ComfyUI + Gradio

프롬프트/레퍼런스 이미지 업로드 → 컨셉보드 자동 완성

2026-1 AI & Interior Architecture Design

Kim Ji-hyun / 2021153021



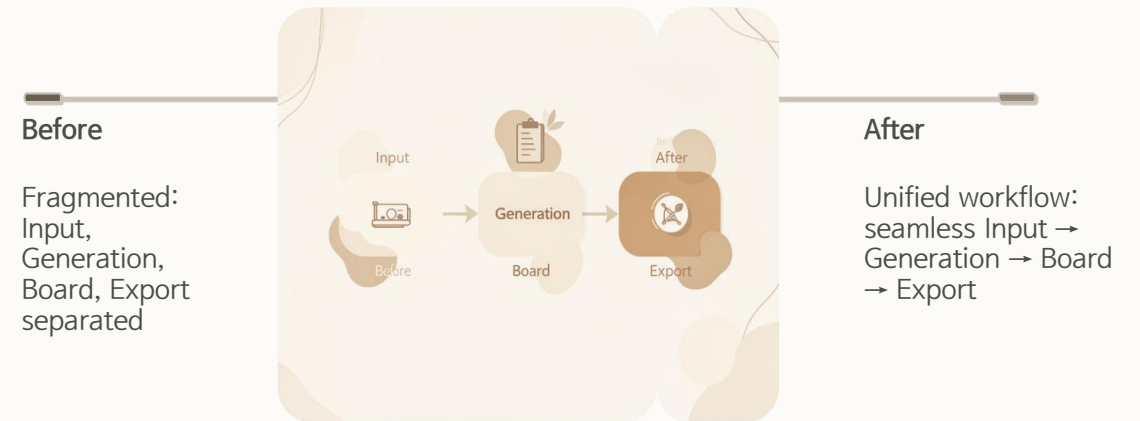
Why This Project?

인테리어 디자인 초기 컨셉 단계의 문제:

- 레퍼런스 이미지 수집 → Pinterest, Google 등 여러 툴
- 재료/조명/무드 정리 → 수작업 메모
- 컨셉 이미지 제작 → Midjourney, Photoshop 별도 작업
- 보드 편집 → Figma, Canva 별도 작업

→ 초기 컨셉 하나 만드는 데 수 시간 소요

→ Fragmented process to a unified process



What I Built

- Claude Code + Gradio + Gemini API + ComfyUI
- Input → AI Analysis → Image Generation → Concept Board → Export
- Fully functional web prototype

Input

1

Space type, mood, material, lighting, activity 선택

Reference image 업로드 → Gemini Vision 자동 분석

Custom keywords 직접 입력 가능

2

AI Analysis

Gemini 2.5 Flash API로 JSON 요청

공간 특성 기반 Design Intent 문장 생성

Main View / Material Detail / Atmosphere 프롬프트
3종 자동 생성

4

Concept Board

생성 이미지 + Design Intent + 컬러 팔레트 자동 구성

재료 스와치 / 태그 / 한 줄 선언문 포함

레퍼런스 이미지 섹션 자동 삽입

Image Generation

3

ComfyUI + FLUX 모델로 이미지 3장 생성

Reference img2img: 업로드 이미지 구조 반영

4x-UltraSharp 업스케일로 고해상도 출력

Export & Management

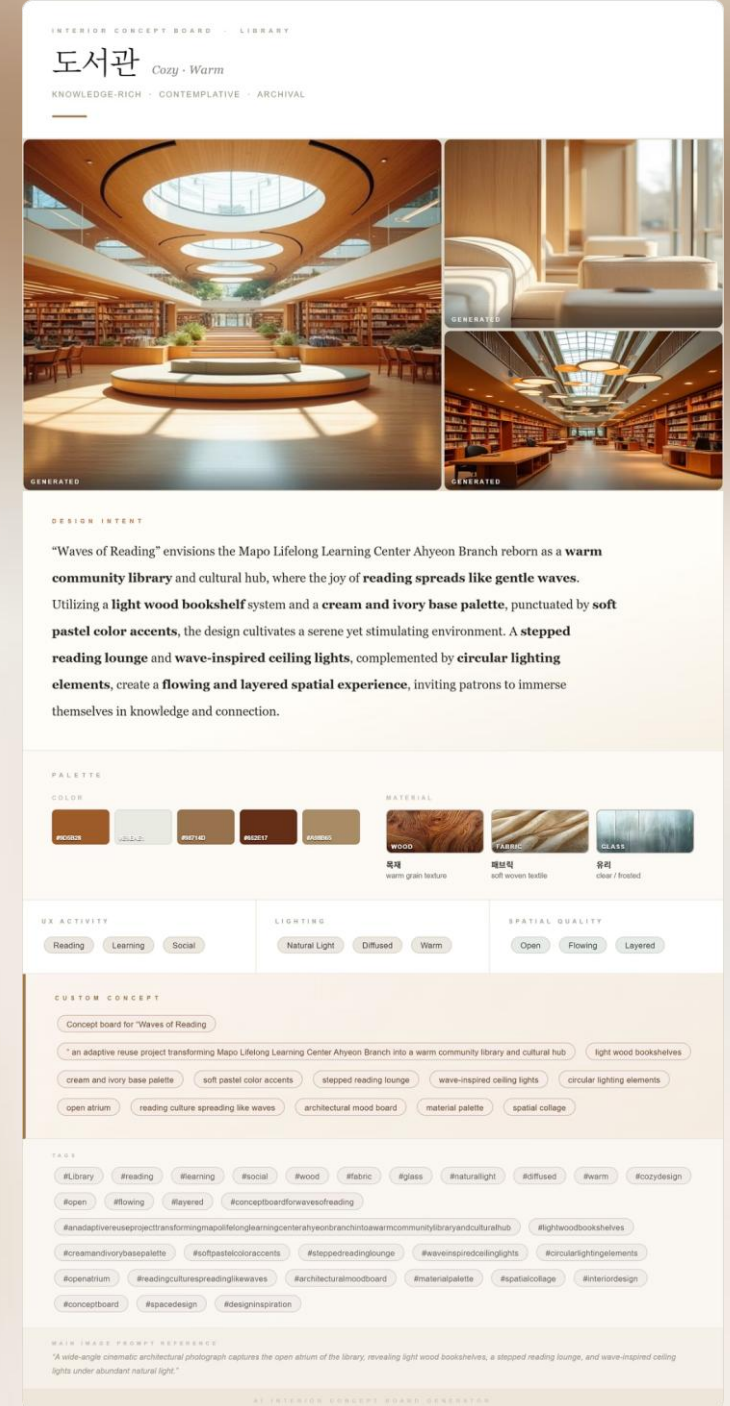
5

PNG / PDF 형식으로 저장

Pin으로 즐겨찾기 저장

History에서 이전 생성 결과 재열람 가능

Admin Panel: API 사용량, 시스템 상태 모니터링



Development with Claude Code

Python 1,670 lines / Gradio 6.x / Implemented with Claude Code

Key Development Challenges & Solutions:

- 1. Port Conflict Resolution: Gradio 기본 포트 충돌 → 동적 포트 할당 구현
- 2. Admin Panel Integration: 사용자 기능과 관리자 기능 분리 → 별도 라우트 구성
- 3. API Error Handling: Gemini/ComfyUI 타임아웃 → Fallback 메커니즘 추가
- 4. Image Processing Pipeline: 메모리 누수 → 배치 처리 최적화

Claude Code의 역할:

- 기능 요청 → 즉시 구현 → 버그 수정 반복
- 복잡한 API 통합 자동화
- 성능 최적화 제안 및 구현

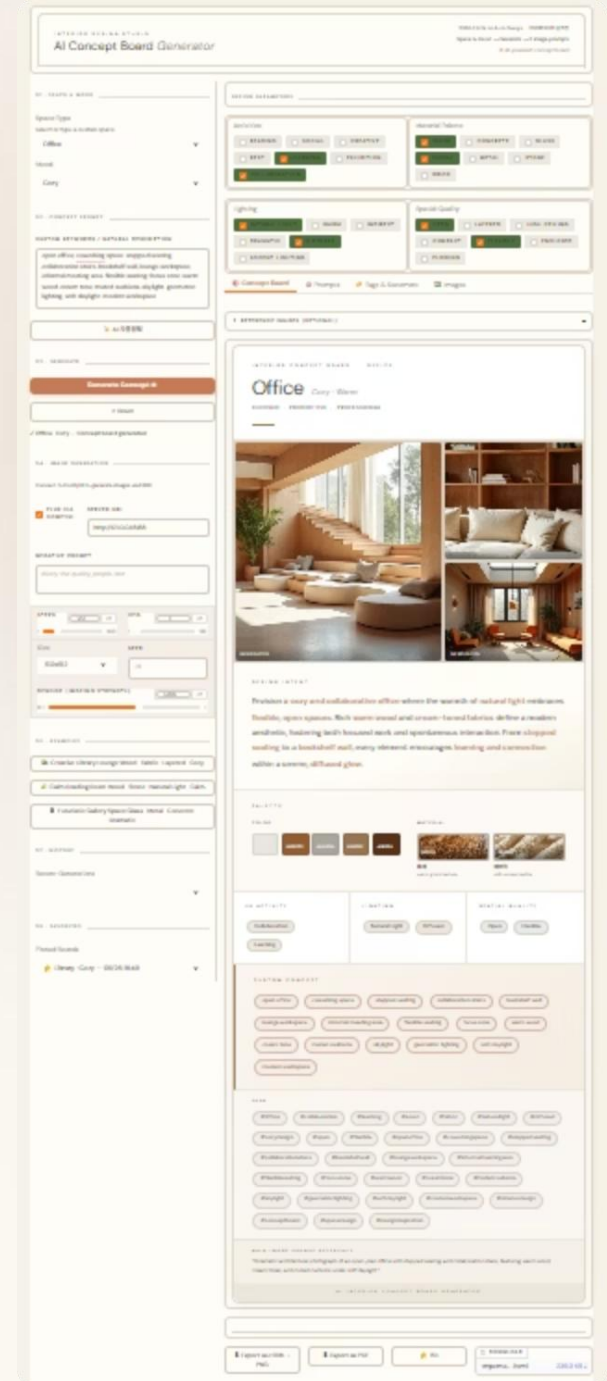
Python	Gradio	Claude Code
1,670 lines	6.x	Implementation tool

```
C:\Program Files\WindowsAp  ×  +  ▾
[ Gemini ] API key loaded: yes
* Running on local URL:  http://0.0.0.0:7860
* Running on public URL: https://eff4299152e3b6f7db.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in
the working directory to deploy to Hugging Face Spaces (https://huggingface.co/spaces)
```

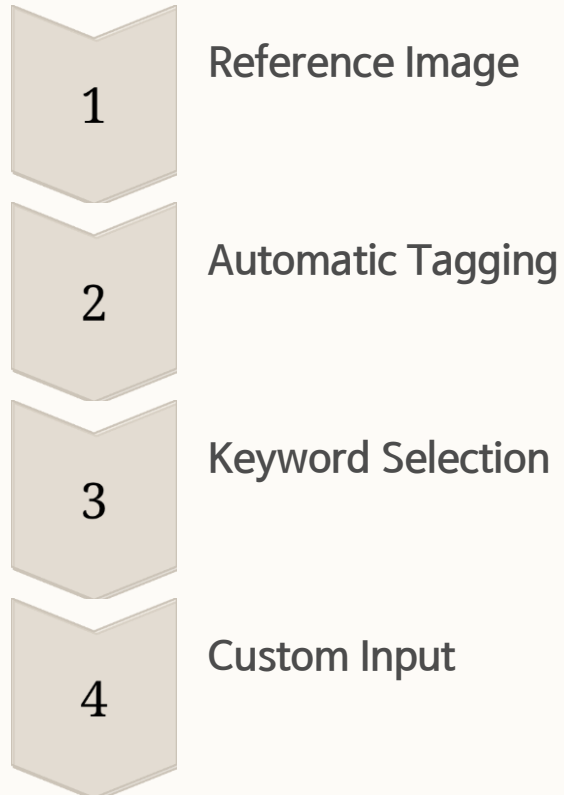
Full Workflow

- 1 — Reference Image Upload
- 2 — Gemini Vision automatic tagging
materials / lighting / mood
- 3 — Keyword selection + Custom Keywords input
- 4 — Click Generate Concept
- 5 — Gemini API → Design Intent + 3 prompt generation
- 6 — ComfyUI FLUX → 3 images
Main / Material / Atmosphere
- 7 — Automatic concept board composition
- 8 — Image Upscale
4x-UltraSharp: 이미지 내용 유지하며 해상도 4배 향상
- 9 — Pin save / History check
- 10 — PNG / PDF Export



Web Interface — Input Stage

Reference Image Upload → Automatic Tagging → Keyword Selection



INTERIOR DESIGN STUDIO

AI Concept Board Generator

2026-I AI & Int.Arch. Design · 2021I53021 김지현
Space & mood → Generate → 3 Image prompts
* AI-powered concept board

01 · SPACE & MOOD

Space Type
Select or type a custom space

Office

Mood

Cozy

02 · CONCEPT PROMPT

CUSTOM KEYWORDS / NATURAL DESCRIPTION

open office, coworking space, stepped seating, collaboration stairs, bookshelf wall, lounge workspace, informal meeting area, flexible seating, focus zone, warm wood, cream tone, muted cushions, skylight, geometric lighting, soft daylight, modern workspace

AI 자동입력

03 · GENERATE

Generate Concept *

Reset

DESIGN PARAMETERS

Activities

☐ READING ☐ SOCIAL ☐ CREATIVE

☐ REST ☒ LEARNING ☐ EXHIBITION

☒ COLLABORATION

Material Palette

☒ WOOD ☐ CONCRETE ☐ GLASS

☒ FABRIC ☐ METAL ☐ STONE

☐ BRICK

Lighting

☒ NATURAL LIGHT ☐ WARM ☐ INDIRECT

☐ DRAMATIC ☒ DIFFUSED

☐ ACCENT LIGHTING

Spatial Quality

☒ OPEN ☐ LAYERED ☐ HIGH CEILING

☐ COMPACT ☒ FLEXIBLE ☐ ENCLOSED

☐ FLOWING

Concept Board Prompts Tags & Statement Images

REFERENCE IMAGES (OPTIONAL)

Upload reference photos — leave empty to auto-generate via AI. Each slot has a role: Main view, Material detail, Atmosphere.

SLOT 1 - MAIN VIEW

이미지를 여기에 드롭
- 또는 -
클릭하여 업로드

SLOT 2 - MATERIAL DETAIL

이미지를 여기에 드롭
- 또는 -
클릭하여 업로드

SLOT 3 - ATMOSPHERE

이미지를 여기에 드롭
- 또는 -
클릭하여 업로드

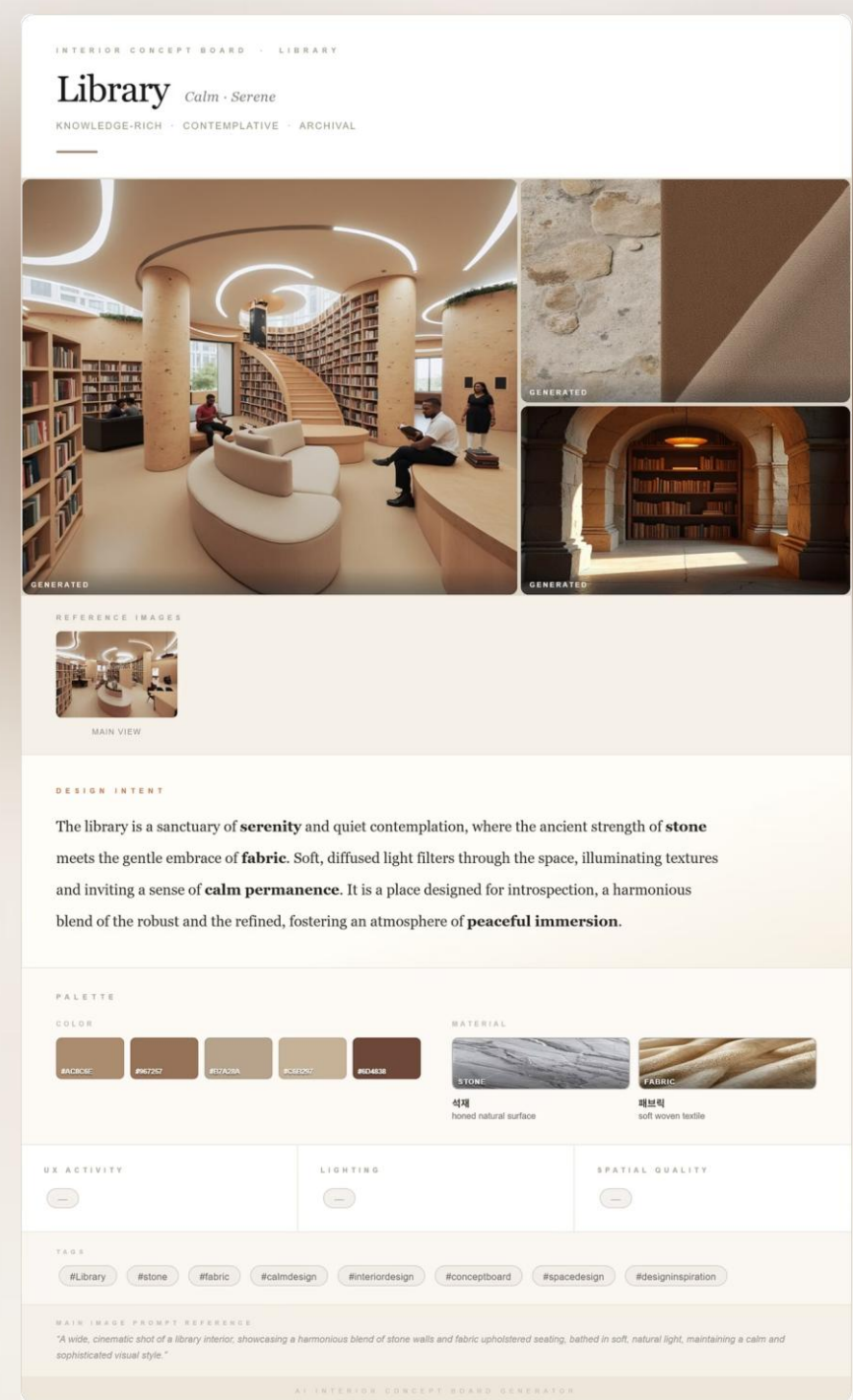
Concept Board — Auto-generated Output

컨셉보드 구성 요소

- ① Space & Mood 공간명 + 무드 태그 (Calm / Serene 등)
- ② Design Intent Gemini API가 생성한 공간 컨셉 서술문
- ③ AI Generated Images
 - Main View — 공간 전체 구도
 - Material Detail — 재료·텍스처 클로즈업
 - Atmosphere — 조명·분위기
- ④ Color Palette 생성 이미지에서 자동 추출한 주요 색상
- ⑤ Material Swatches 선택한 재료 시각화 (Wood / Concrete / Fabric 등)
- ⑥ Tags & Statement AI 분석 키워드 + 한 줄 디자인 선언문
- ⑦ Reference Images 업로드한 레퍼런스 이미지 표시

모든 요소는 Generate Concept 한 번으로 자동 구성됨

다양한 공간 유형(주거, 상업, 공공)에서도 일관된 품질의 컨셉보드 생성 가능



Diverse Space Types — Consistent Quality

Office / Pop-up Store / Children's Hospital / Library — 다양한 공간 유형에서 생성된 컨셉보드 예시

INTERIOR CONCEPT BOARD

Office

Calm · Serene

FOCUSED · PRODUCTIVE · PROFESSIONAL



DESIGN INTENT

Inspired by the serene embrace of a forest canopy, this office design cultivates a **calm and creative** sanctuary. **Layered wooden ceilings** and **tree-like columns** evoke nature's grandeur, while **soft, diffused natural light** permeates the **flowing spaces**. A palette of **warm beige and olive**, accented by **biophilic elements** and **green lounge islands**, fosters a harmonious environment for **collaboration** and focused work.

PALETTE

COLOR

#C8A28E

#D9C7A8

#A68B5B

#E8DCC8

#A68B5B

MATERIAL

WOOD

TEXTILE

목재

나무 grain texture

천/커튼

soft woven textile

UX ACTIVITY

Creative

Collaborative

LIGHTING

Natural Light

Warm

Diffused

SPATIAL QUALITY

Layered

Flowing

CUSTOM CONCEPT

Biophilic office interior inspired by a forest canopy

tree-like columns

layered wooden ceiling

soft natural lighting

green lounge islands

curved workstations

indoor plants

warm beige and olive color palette

calm creative atmosphere

realistic architectural visualization

TAGS

#Office

#creative

#collaborative

#peace

#biophilic

#natural light

#warm

#diffused

#interior design

#layered

#flowing

#biophilic office interior

#layered wooden ceiling

#tree-like columns

#layered wooden ceiling

#soft natural lighting

#green lounge islands

#curved workstations

#indoor plants

#warm beige and olive color palette

#calm creative atmosphere

#realistic architectural visualization

#interior design

#concept board

#space design

#design inspiration

MAIN IMAGE PROMPT RESEARCH

"Cinematic architectural photograph of a biophilic office interior. Featuring tree-like columns, layered wooden ceiling, natural light filtering through, indoor plants, warm beige and olive color palette, soft natural light, and a calm creative atmosphere."

3D INTERIOR CONCEPT BOARD GENERATED

INTERIOR CONCEPT BOARD

Pop-up Store

Dynamic · Energetic

CONTEMPORARY INTERIOR ARCHITECTURE



DESIGN INTENT

Envision a dynamic pop-up store where **modular display blocks** and **colorful curved partitions** orchestrate a playful yet sophisticated interior. This space, designed for both **exhibition** and **social interaction**, embraces **flexible and open spatiality**. **Bold accent colors** punctuate a warm neutral base, while a **playful lighting installation** with accent and diffused elements cultivates an energetic and interactive atmosphere. The design is temporary yet refined, promising a memorable and visually striking experience.

PALETTE

COLOR

#C8A28E

#D9C7A8

#A68B5B

#E8DCC8

#A68B5B

MATERIAL

WOOD

TEXTILE

목재

나무 grain texture

천/커튼

soft woven textile

UX ACTIVITY

Exhibition

Social

LIGHTING

Accent Lighting

Diffused

SPATIAL QUALITY

Flexible

Open

CUSTOM CONCEPT

Experimental pop-up store interior

modular display blocks

colorful curved partitions

playful lighting installation

interactive product zones

bold accent colors with warm neutral base

temporary but sophisticated spatial design

energetic atmosphere

realistic interior render

TAGS

#Pop-up store

#exhibition

#social

#flexible

#modern

#reconfigurable

#dynamic lighting

#bold

#modern design

#flexible

#open

#experimental pop-up store interior

#modular display blocks

#colorful curved partitions

#playful lighting installation

#interactive product zones

#bold accent colors with warm neutral base

#temporary but sophisticated spatial design

#energetic atmosphere

#realistic interior render

#interior design

#concept board

#space design

#design inspiration

MAIN IMAGE PROMPT RESEARCH

"A wide, cinematic architectural photograph of an experimental pop-up store interior, showcasing modular display blocks and colorful curved partitions under a playful lighting installation, rendered with a consistent bold and warm neutral color palette."

3D INTERIOR CONCEPT BOARD GENERATED

INTERIOR CONCEPT BOARD

Children's Hospital

Calm · Serene

CONTEMPORARY INTERIOR ARCHITECTURE



DESIGN INTENT

This children's hospital interior is a serene and imaginative **space-themed clinic**, where the gentle glow of **diffused, warm, indirect lighting** from **floating, planet-like pendant lights** illuminates a playful planetary orbit ceiling. The design prioritizes a **calm yet fun atmosphere** through **flowing, open spatial arrangements**, a **central round waiting island**, and **curved soft seating**, all rendered in a soothing palette of **pastel blue, lavender, mint, and beige**. Emphasizing both **learning and rest**, the environment integrates **warm wood textures** and **safe, rounded furniture** to create a friendly, clean, and imaginative healthcare space.

PALETTE

COLOR

#C8A28E

#D9C7A8

#A68B5B

#E8DCC8

#A68B5B

MATERIAL

WOOD

TEXTILE

목재

나무 grain texture

천/커튼

soft woven textile

UX ACTIVITY

Learning

Rest

LIGHTING

Diffused

Warm

Indirect

SPATIAL QUALITY

Flowing

Open

CUSTOM CONCEPT

Space-themed children's clinic interior

playful planetary orbit ceiling

floating planet-like pendant lights

central round waiting island

curved soft seating

warm wood texture

pastel blue

lavender

mint and beige palette

friendly and imaginative healthcare space

safe rounded furniture

calming but fun atmosphere

clean hospital interior

realistic architectural visualization

TAGS

#Children's hospital

#learning

#rest

#wood

#blue

#diffused

#warm

#indirect

#interior design

#flowing

#open

#space-themed children's clinic interior

#playful planetary orbit ceiling

#floating planet-like pendant lights

#central round waiting island

#curved soft seating

#warm wood texture

#pastel blue

#lavender

#mint and beige palette

#friendly and imaginative healthcare space

#safe rounded furniture

#calming but fun atmosphere

#clean hospital interior

#realistic architectural visualization

#interior design

#concept board

#space design

#design inspiration

MAIN IMAGE PROMPT RESEARCH

"Cinematic architectural photograph of a children's hospital main waiting area featuring a space-themed interior with a playful planetary orbit ceiling, floating planet-like pendant lights, a central round waiting island, and curved soft seating, rendered in a pastel blue, lavender, mint, and beige palette with warm wood textures, creating a friendly and imaginative healthcare space."

3D INTERIOR CONCEPT BOARD GENERATED

INTERIOR CONCEPT BOARD

Library

Calm · Serene

KNOWLEDGE-RICH · CONTEMPLATIVE · ARCHIVAL



DESIGN INTENT

Envisioning a **warm public library interior**, this design cultivates a **calm and inviting atmosphere** for lifelong learning. The space embraces an **open and flexible layout** with a **high ceiling**, encouraging both individual contemplation and vibrant community interaction. Integrated **wooden bookshelf walls** and a **stepped reading amphitheater** define a unique architectural structure, where books themselves become a key design element, all bathed in **warm natural light**.

PALETTE

COLOR

#C8A28E

#D9C7A8

#A68B5B

#E8DCC8

#A68B5B

MATERIAL

WOOD

TEXTILE

목재

나무 grain texture

천/커튼

soft woven textile

UX ACTIVITY

Reading

Learning

Social

LIGHTING

Natural Light

Warm

Diffused

SPATIAL QUALITY

Open

Flexible

High Ceiling

CUSTOM CONCEPT

warm public library interior

community library reading hall

stepped reading amphitheater

book stairs

floor-to-ceiling bookshelves

integrated wooden bookshelf walls

books as architectural structure

open reading plaza

cultural community space

flexible learning library

flexible seating area

informal reading lounge

lecture and reading steps

book nook space

light birch wood

cream-colored platforms

soft cushioned seating blocks

neutral red

olive green

dusty blue

beige cushions

calm and inviting atmosphere

warm natural light

large skylight ceiling

geometric ceiling lights

warm ambient lighting

soft shadows

modern public library

Scandinavian wooden interior

architectural interior render

high-quality architectural visualization

TAGS

#Library

#reading

#learning

#social

#wood

#blue

#natural light

#warm

#diffused

#interior design

#open

#flexible

#high ceiling

#warm public library interior

#community library reading hall

#stepped reading amphitheater

#book stairs

#floor-to-ceiling bookshelves

#integrated wooden bookshelf walls

#books as architectural structure

#open reading plaza

#cultural community space

#flexible learning library

#flexible seating area

#informal reading lounge

#lecture and reading steps

#book nook space

#light birch wood

#cream-colored platforms

#soft cushioned seating blocks

#neutral red

#olive green

#dusty blue

#beige cushions

#calm and inviting atmosphere

#warm natural light

#large skylight ceiling

#geometric ceiling lights

#warm ambient lighting

#soft shadows

#modern public library

#Scandinavian wooden interior

#architectural interior render

#high-quality architectural visualization

#interior design

#concept board

#space design

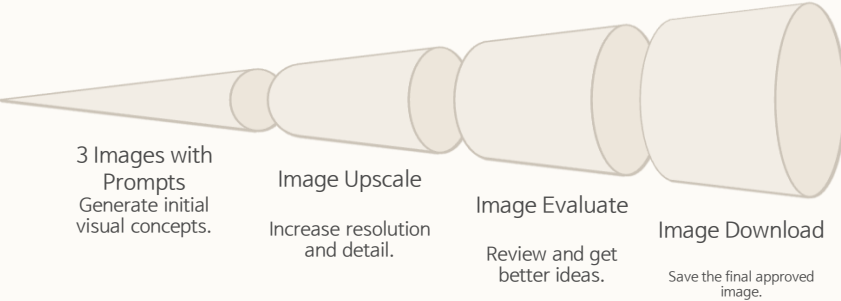
#design inspiration

MAIN IMAGE PROMPT RESEARCH

"Cinematic architectural photograph of a modern public library, showcasing a stepped reading amphitheater with floor-to-ceiling light wood bookshelves, flexible seating blocks in muted red, olive green, dusty blue, and beige, under a large skylight ceiling, evoking a contemplative academic interior space."

3D INTERIOR CONCEPT BOARD GENERATED

Image Upscale & Evaluation



Concept Board

Prompts

Tags & Statement

Images

Hashtags

#Lounge #social #rest #wood #fabric #warm #diffused #cozydesign #open #flowing #pethoteloungeinterior #curvedpetrestingpods #softroundedfurniture #warmwoodfloor #playfulpastelaccents #cleanmodernpetcarespace #cozyandfunatmosphere #realisticarchitecturalrender #interiordesign #conceptboard #spacedesign #designinspiration

The Pet Hotel Lounge is a haven of **cozy sophistication**, designed to nurture both pets and their owners. Warm wood floors invite playful paws, while **soft, rounded furniture** and **curved pet resting pods** create an atmosphere of serene comfort. Playful pastel accents dance within a **clean, modern pet-care space**, ensuring a **fun and flowing** environment that is both aesthetically pleasing and incredibly functional.

Concept Board

Prompts

Tags & Statement

Images

SLOT 1 — MAIN CONCEPT

Cinematic architectural photography of a pet hotel lounge, featuring warm wood floors, soft rounded furniture, curved pet resting pods, playful pastel accents, and an open, flowing spatial design.

SLOT 2 — MATERIAL / DETAIL

Close-up of a pet hotel lounge interior, showcasing the warm wood grain, soft fabric textures of the rounded furniture, and the smooth, matte finish of the pastel accents.

SLOT 3 — ATMOSPHERE

Atmospheric shot of the pet hotel lounge at dusk, bathed in warm, diffused lighting that highlights the cozy mood and the playful pastel accents against the soft, rounded forms.

Concept Board

Prompts

Tags & Statement

Images

Generated Images — select size and re-generate for higher quality.

MAIN VIEW

Size: 1024x1024

Re-generate Hi-Res

MATERIAL DETAIL

Size: 1024x1024

Re-generate Hi-Res

Evaluate Images (Gemini)

Main View: Score: 7/10 Assessment: The image successfully conveys a cozy and inviting mood through its soft color palette and comfortable seating, but the materiality feels somewhat flat. Improvements:
1. Add subtle texture to the walls and upholstery to enhance the sense of warmth and depth.
2. Introduce more varied lighting, perhaps with some softer, diffused sources, to create more intimate pockets within the lounge.

Material Detail: Score: 7/10 Assessment: The image effectively conveys a soft, inviting mood through the use of plush, rounded furniture in a warm pink hue, suggesting comfort and a cozy lounge atmosphere. Improvements:
1. Introduce a soft throw blanket or a decorative pillow with a subtle texture on one of the chairs to enhance the sense of warmth and tactile comfort.
2. Include a warm, diffused light source, such as a table lamp with a fabric shade, in the foreground or midground to further emphasize the cozy ambiance.

Atmosphere: Score: 7/10 Assessment: The image effectively conveys a cozy lounge atmosphere through its soft lighting and plush, rounded furniture, though the materiality feels somewhat flat. Improvements:
1. Add more varied textures like a chunky knit throw blanket or a velvet cushion to enhance the sense of comfort and luxury.
2. Introduce subtle natural elements, such as a small potted plant or a wooden decorative bowl, to add warmth and organic contrast.

Concept Board

Prompts

Tags & Statement

Images

REFERENCE IMAGES (OPTIONAL)

INTERIOR CONCEPT BOARD — LOUNGE

Lounge Cozy · Warm

RELAXED · SOCIAL · COMFORTABLE

DESIGN INTENT

The Pet Hotel Lounge is a haven of **cozy sophistication**, designed to nurture both pets and their owners. Warm wood floors invite playful paws, while **soft, rounded furniture** and **curved pet resting pods** create an atmosphere of serene comfort. Playful pastel accents dance within a **clean, modern pet-care space**, ensuring a **fun and flowing** environment that is both aesthetically pleasing and incredibly functional.

COLOR

COZY

WARM

SOFT

MODERN

PLAYFUL

MATERIAL

WOOD

织物

warm grain texture

FABRIC

棉麻

soft woven textile

USE ACTIVITY

Social

Rest

LIGHTING

Warm

Diffused

SPATIAL QUALITY

Open

Flowing

CUSTOM CONCEPT

Pet hotel lounge interior

curved pet resting pods

soft rounded furniture

warm wood floor

playful pastel accents

clean modern pet-care space

cozy and fun atmosphere

realistic architectural render

TAGS

#lounge

#social

#rest

#wood

#fabric

#warm

#diffused

#cozydesign

#open

#flowing

#pethoteloungeinterior

#curvedpetrestingpods

#softroundedfurniture

Made with GAMMA

Export & Management

User Features (Pin Save / History / Export) + Administration Panel (API Usage / System Status)

01

Pin Save

03

PNG / PDF Export

Save & Organize

- Pin favorite concept boards
- Access full history of generated designs
- Quick retrieval of previous work

Export Options

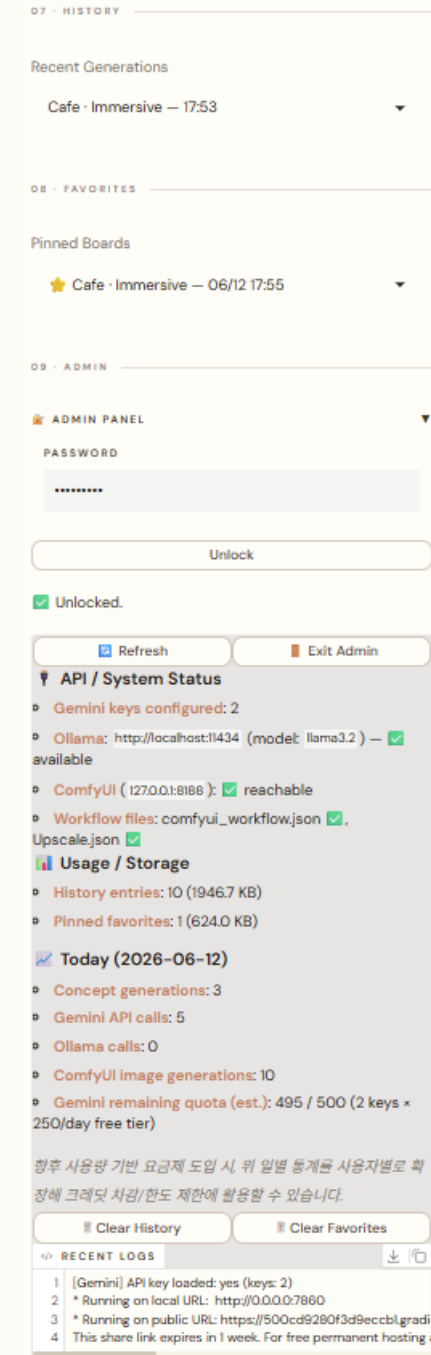
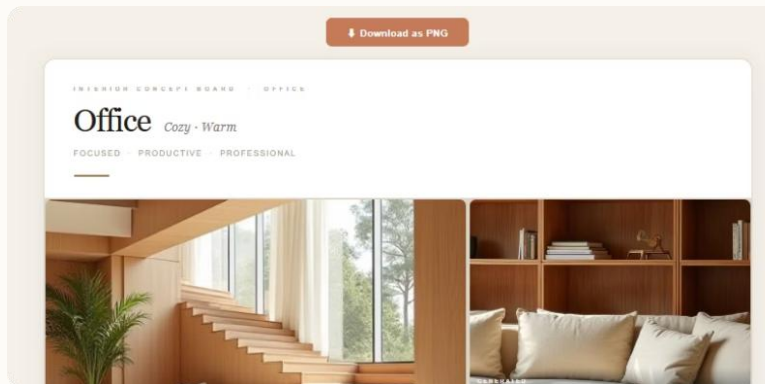
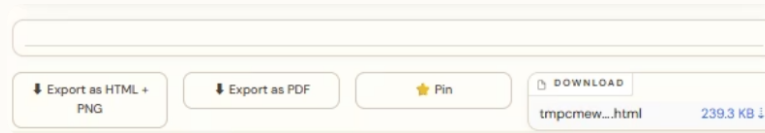
- PNG export for web and presentations
- PDF export for client deliverables
- High-resolution output ready for production

02

History / Favorites

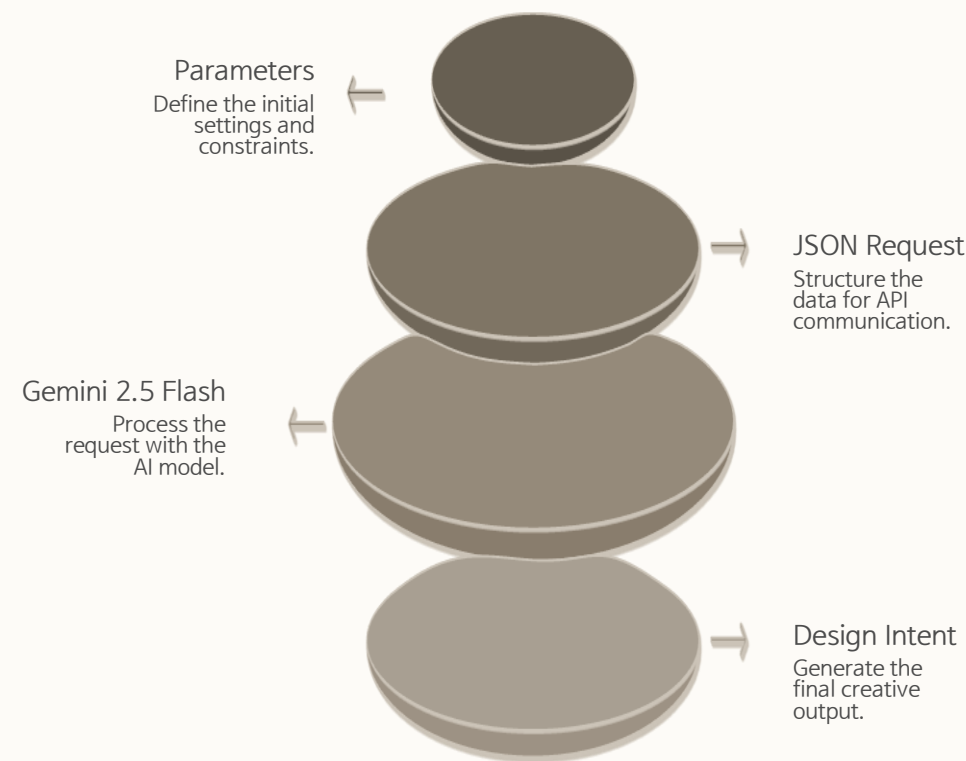
04

Administration Panel



Gemini API — Concept Generation

This visual shows how parameters are assembled into a JSON request, sent to Gemini 2.5 Flash, and returned as design intent with three prompts.



01 · SPACE & MOOD

Space Type
Select or type a custom space

Office

Mood

Cozy

02 · CONCEPT PROMPT

CUSTOM KEYWORDS / NATURAL DESCRIPTION

open office, coworking space, stepped seating, collaboration stairs, bookshelf wall, lounge workspace, informal meeting area, flexible seating, focus zone, warm wood, cream tone, muted cushions, skylight, geometric lighting, soft daylight, modern workspace

AI 자동입력

03 · GENERATE

Generate Concept *

Reset

✓ Office · Cozy — Concept board generated

04 · IMAGE GENERATION

Connect to ComfyUI to generate images via FLUX

DESIGN PARAMETERS

Activities

READING

SOCIAL

CREATIVE

REST

LEARNING

EXHIBITION

COLLABORATION

Material Palette

WOOD

CONCRETE

GLASS

FABRIC

METAL

STONE

BRICK

Lighting

NATURAL LIGHT

WARM

INDIRECT

DRAMATIC

DIFFUSED

ACCENT LIGHTING

Spatial Quality

OPEN

LAYERED

HIGH CEILING

COMPACT

FLEXIBLE

ENCLOSED

FLOWING

Concept Board

Prompts

Tags & Statement

Images

SLOT 1 — MAIN CONCEPT

Cinematic architectural photograph of an open-plan office with stepped seating and collaboration stairs, featuring warm wood, cream tones, and muted cushions under soft daylight.

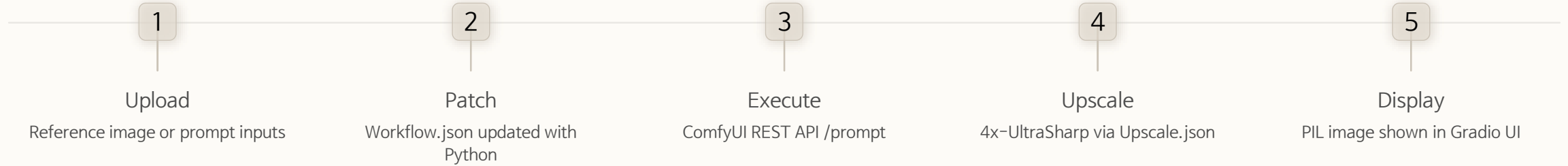
SLOT 2 — MATERIAL / DETAIL

Close-up detail of a bookshelf wall crafted from warm wood, showcasing the texture of cream-toned fabric on flexible seating, under a diffused skylight.

SLOT 3 — ATMOSPHERE

An atmospheric shot of an informal meeting area bathed in soft daylight from a skylight, highlighting the cozy mood with geometric lighting fixtures and warm wood accents.

ComfyUI — Image Generation Pipeline



워크플로우 구성

- FLUX 모델 기반 workflow.json을 Python으로 런타임 동적 패치
- ComfyUI REST API(/prompt)로 전송 실행

동적 패치 항목

- 프롬프트 (Positive / Negative)
- Seed / Steps / CFG / Denoise / 해상도

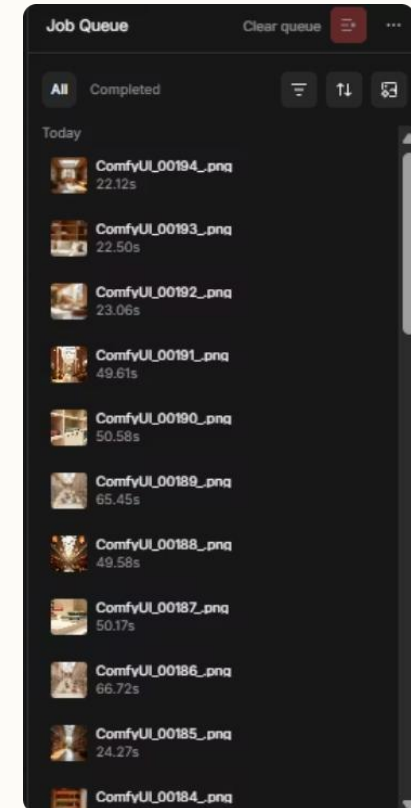
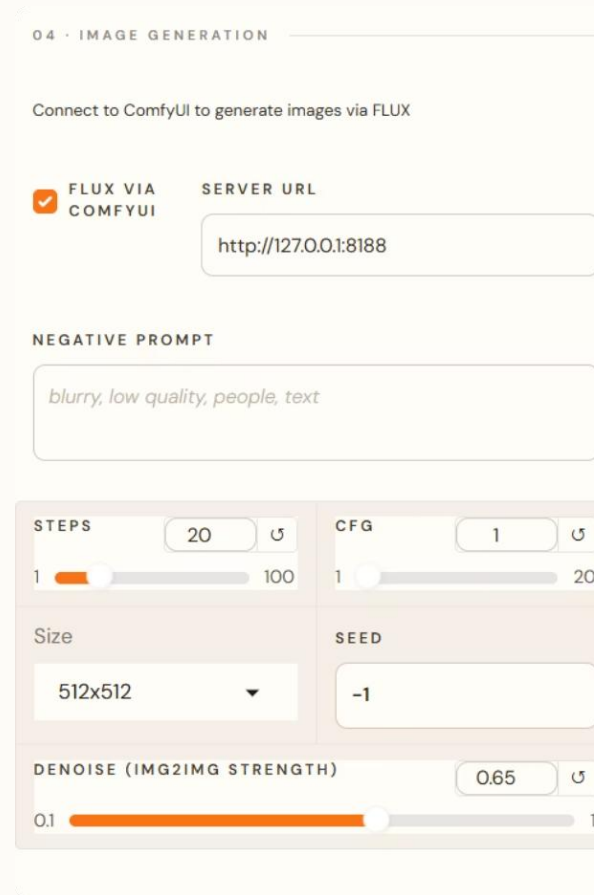
생성 이미지 3종 Main View / Material Detail / Atmosphere

업스케일 4x-UltraSharp 모델로 고해상도 변환

Fallback ComfyUI 미연결 시 Pollinations.ai(무료 외부 API)로 자동 전환

img2img 모드

- 레퍼런스 이미지 업로드 → LoadImage + VAEEncode 노드 동적 삽입
- KSampler latent_image를 VAEEncode 출력으로 교체
- Denoise: 낮을수록 원본 유사 / 높을수록 새로운 디자인



AI Interior Concept Board Generator



The system connects initial research and AI-assisted development to a collaborative final output.

Meaning

AI is not replacing design, but rather a supporting tool that rapidly connects the designer's initial concept visualization process.

Implementation Stack

Gradio UI / Gemini API / ComfyUI FLUX / img2img / Upscale / Export

📄 Key Achievements

- Automated concept board generation from reference images
- Real-time AI-powered design parameter extraction
- Seamless workflow from input to export